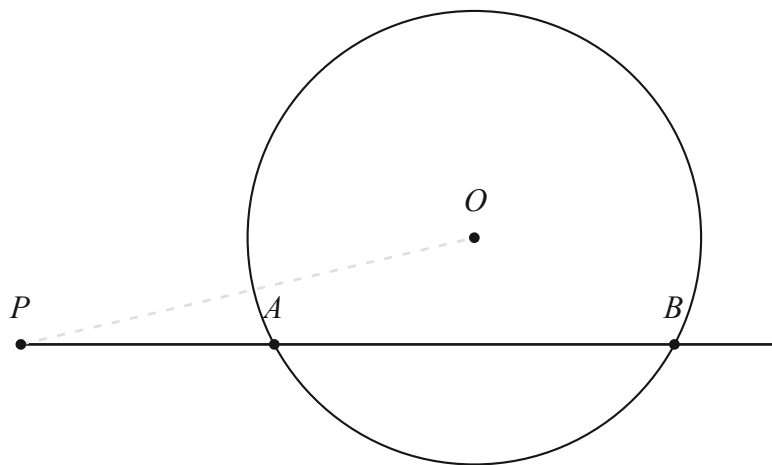


The circle geometry toolkit

For a circle with centre O and radius r , the signed power of a point P is $PO^2 - r^2$. One quantity unifies secants, tangents, and intersecting chords.



Exterior secant. If a ray from an exterior point P meets the circle first at A and then at B , then $PA \cdot PB = PO^2 - r^2$.

Tangent. If PT is tangent at T , then $PT^2 = PO^2 - r^2$. The radius OT is perpendicular to PT .

Interior chords. If chords AB and CD cross at an interior point P , then $PA \cdot PB = PC \cdot PD = r^2 - PO^2$, using positive lengths.

A quadratic remembers both intersections

Put P at the origin. Let u be a unit vector along the secant, and let o be the position vector of the centre. A point tu lies on the circle exactly when

$$|tu - o|^2 = r^2.$$

Expanding gives a quadratic in the signed line coordinate t :

$$t^2 - 2(u \cdot o)t + |o|^2 - r^2 = 0.$$

By the product-of-roots identity, the two intersection coordinates have product $|o|^2 - r^2$. For an exterior point and a secant ray, both roots are positive, giving $PA \cdot PB$.

The radical axis

For two circles, write the power of X to each and equate the expressions. The quadratic $X \cdot X$ terms cancel, leaving a linear equation when the centres differ. The resulting line is called the radical axis.

Example: compare $x^2 + y^2 = 25$ and $(x - 6)^2 + y^2 = 9$. Equality of powers gives

$$x^2 + y^2 - 25 = (x - 6)^2 + y^2 - 9,$$

so $12x = 52$, or $x = 13/3$. The radical axis is perpendicular to the horizontal line joining the centres.

For concentric circles of different radii there is no equal-power locus; for identical circles every point has equal power.

Get the lengths in the right order

Worked example. From an exterior point, the near intersection is 3 units away and the far intersection is 12 units away. The power is 36, so a tangent from that point has length 6.

The product uses the whole distance to the far intersection. The segment inside the circle has length 9, but multiplying 3 by 9 would give the wrong answer.

Exercises

1. A tangent from P has length 8. A secant meets the circle first 4 units from P. How far from P is the second intersection?
2. Chords cross at P inside a circle. One has segment lengths 2 and 9; the other has one segment of length 3. Find its remaining segment.
3. A circle has radius 5, and $PO = 3$. Find the signed power of P and the positive product of the segments of any chord through P.

Solutions

1. The power is $8^2 = 64$. Thus $4 \cdot PB = 64$, giving $PB = 16$. The portion inside the circle is 12.
2. Equal products give $2 \cdot 9 = 3x$, so $x = 6$.
3. The signed power is $3^2 - 5^2 = -16$. The positive chord-segment product is 16.

Habit to keep: decide whether you are using signed coordinates or positive segment lengths before writing a power equation.